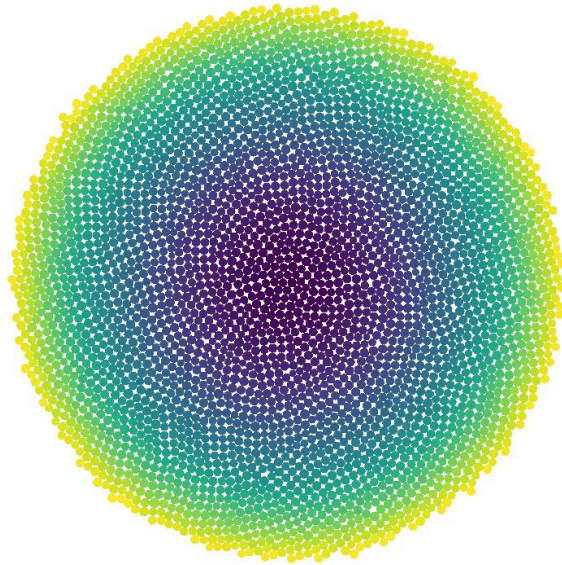


Methods & Data



Amin Sabri Al-Ait

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Prologue

From here on out, we will address the potential put by the game on players as “claims”. That is, the game *claims* that player X will reach this potential.

Generally, this study will take John Doe's overall rise and potential breaking as inspiration to study a sample of players from each game in the EA FIFA series to see if we can take EA FIFA as a good and reliable resource for predicting future stars.

Specifically, and recalling that EA FIFA has annual releases of the game, so we will assume that every year from 2007 till 2021 will refer to the particular EA FIFA game release of that particular year. In addition, the terms “the game”, “EA FIFA”, “EA FIFA release”, “installments”, “versions” and “year” will be used interchangeably, but they mean the same thing.

Each EA FIFA release gets numerous updates. These updates are usually for bug fixes but most importantly to keep the players’ stats as updated to real time changes as possible. Thus, it is important to note that, for the sake of consistency, only the first “factory” version of each annual EA FIFA release is scraped to build the data set.

The study will focus on a sample of players from 2007 till, and including, 2021. Moreover, the players of each year will be of age 19. And that is to restrict the data to players that are still considered youth (teenagers), and have an uncertain future, thus an uncertain overall score*.

The Data

There are 5,400 players, each is tracked from the year he was 19 years old in, till the year 2021, this selection is displayed best in figure 1.

The 360 19 year old players of every year were chosen after sorting all the 19 year old players of a particular year by highest to lowest potential, and then scraping the top 360.

The potential range, as shown in figure 2, spans from 64 till 94.

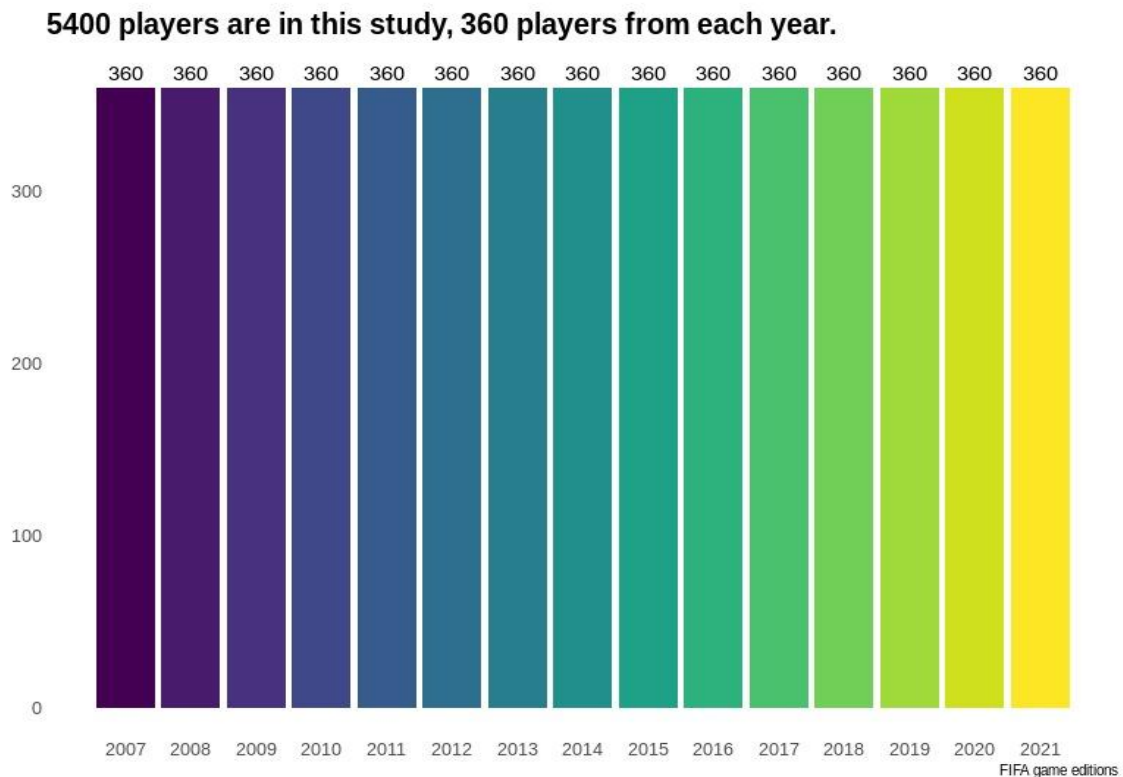


Figure 1

Data Scraping

Data has been scraped using R packages **rvest** (which you can find it on [github](#), and read about it [here](#)) and **rselenium** (which you can find it on [github](#), and read about it [here](#)).

sofifa.com

The data is scraped from <https://sofifa.com>. Every player has the following information scraped in every year:

- Short name
- Long name
- Age
- Nationality
- Club
- Position: in the team formation. For example:

- RW: Right Wing
- CM: Center Midfielder
- CB: Center back
- Etc...
- Overall
- Potential: only scrapes this score when the player is 19 years old
- FifaIndex: which year this player was 19 years old
- Year
- ID (put by SoFifa.com)

transfermarkt.de

Data from this website was scraped only for a specific list of players**. The aim is to get real life performance data on these players.

Every player has the following information scraped:

- Year
- Name
- Competition
- Club
- Games Played
- Goals
- Assists
- Minutes Played

The Keywords

The study will use a few special keywords, some were scraped variables from the players' profiles like **overall** and **potential**, and others are the result of data processing, for example: **prime** and **potential brackets**. These keywords are explained (or re-explained) as the following:

- Overall: a player's score out of 100 in a given year.
- Potential: a score (out of 100) that EA FIFA claims this player will reach.
- Prime: the highest overall achieved by a player in his respective career.
- Potential Brackets: EA FIFA puts potential scores on players which vary a lot from one another, thus they have been divided into three brackets/segments, and one player will fall into one and only one bracket/segment. And they are:
 - [64, 74[
 - [74, 84[

- [84, 94]

Methods

Explanations of data manipulations, data summaries and methods:

1. Intuition behind Potential brackets of figures 1, 3 and 8
 - a. Players' potential scores are spread out, according to figure 2, between the values of 64 and 93. The idea of potential brackets is to break down this spread into 3 segments:
 - i. First bracket starts at the minimum of the spread 64 and ends before 74: **[64, 74[**
 - ii. Second bracket starts at 74 and ends before 84: **[74, 84[**
 - iii. Third bracket starts at 84, and ends at the maximum of the spread 94: **[84, 94]**
2. Method behind measuring each potential bracket's accuracy throughout the years in figure 8
 - a. The overall scores of players of any particular year (360 19 year old players) are tracked throughout their career and always measured against the potential score they had at their age of 19. The correct claims (number of potential achievers) of that specific potential bracket of that specific year are then divided by the total number of players that fill into that potential bracket of that particular year times a 100.

Cover picture is formed of 5,400 tiny circles, each representing a player of our study. Color and radius (although radius difference is not that obvious to the human eye) are determined by the players' prime overall scores. Which means Lionel Messi's circle should be the largest and should be placed at the exact center.

*In general, the older a player the easier it gets to predict how their career (and thus overall) will take shape.

**These players are specific cases that will be remarked on in the text of the research document "How Accurate Is FIFA At Predicting Future Stars?".